modulator consists of an array of proposed smart glass elements known as dual-cell liquid crystal shutters (DLSs). The liquid crystal shutter array works like a filter to encode signals into the light as it passes. It would need just one watt of power to operate, which can be supplied using a small solar panel. This provides a greener mode of communication compared to conventional Wi-Fi or cellular data transmission.

Robotic gripper uses simple inflation to wrap around objects

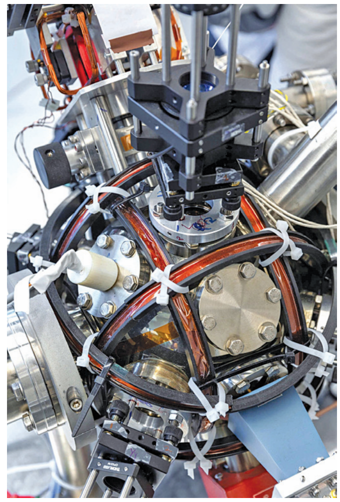
Smooth gripper grasps succulent (Source: https://news8plus.com)

Robotic gripper developed by researchers from Massachusetts uses a collection of thin tentacles to entangle and trap objects, similar to how jellyfish collect stunned prey. The gripper depends on simple inflation to wrap around objects, and there is no need for sensing, planning, or feedback control. The gripper's strength and adaptability come from its ability to entangle itself with the object it is attempting to grasp. The foot-long filaments are hollow rubber tubes. One side of the tube has thicker rubber than the other, so when the tube is pressurised, it curls like a pigtail or like straightened hair on a rainy day.

Automated robotic system keeps restrooms in public places clean

Clean restrooms are very essential for convenience stores to encourage customers to stay longer and provide good business. To maintain consistency in the cleanliness of restrooms, researchers at Tokyo Metropolitan University have developed an automated robotic system for cleaning restrooms in convenience stores and other public spaces. This would simplify the work of shop clerks and convenience store cleaners. The system is configured with a toilet bowl top surface cleaning mechanism, a toilet bowl lifting mechanism, and a restroom floor cleaning robot. By robotising the restroom space itself, the system aims to streamline the cleaning operation and achieve accurate cleaning in a short time (less than 20 seconds).

Quantum based accelerometer for navigation without GPS

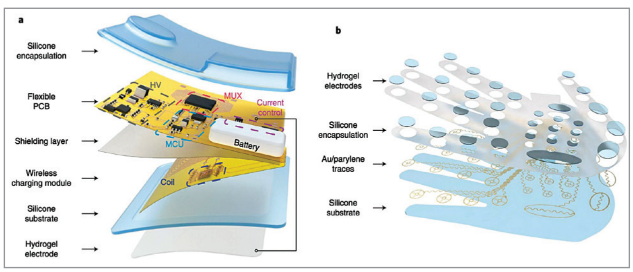


Quantum 'compass' could allow navigation without relying on satellites (Source: https://www.digitaltrends.com)

Navigation without relying on satellites is not possible. Therefore, researchers at Imperial College London have invented a quantum 'compass' that allows navigation without the need for GPS. The newly developed quantum accelerometer has a very high level of accuracy. This is achieved by measuring the movements of super-cool atoms, which are cooled to such a degree that they display quantum behaviour; they are both particles and waves. The wave properties of the atoms are affected by acceleration, so scientists examine the movements of the atoms by creating an atom interferometer—a tool that measures the displacement of waves. This allows the accelerometer to measure movement through space highly accurately.

Ultra-thin system that produces tactile sensations on a user's skin

Researchers from the University of Hong Kong, the City University of Hong Kong, the University of Electronic Science and Technology of China (UESTC), and other institutes in China have invented 'WeTac,' which is a small, soft, and ultra-thin wireless electrotactile system that produces tactile sensations on a user's skin. WeTac consists of a series of electrodes placed over a user's palm and miniaturised soft electronic components acting as the device's control panel. When worn by users, the device can produce



Schematic illustration of the miniaturised wireless electrotactile system. (a) Explosive view of the structures and components of the driver unit, (b) Explosive view of the structure for the hydrogel-based electrodes hand patch (Source: https://techxplore.com)